

RICHLANDS ELLIS, NC, USA

WMO: 723069

Lat: 34.833N

Lon: 77.617W

Elev: 29

StdP: 100.97

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 93753

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.0	-4.1	-22.5	0.5	5.4	-16.3	0.9	4.6	10.6	17.2	9.1	15.8	1.2	20	0.415

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	34.8	25.3	33.3	24.8	32.4	24.4	26.8	32.5	26.1	31.6	25.5	30.5	3.8	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.2	20.4	29.2	24.6	19.6	28.6	24.0	18.9	28.0	83.8	32.5	81.0	31.6	78.1	30.7	30.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.7	7.6	6.8	-10.7	37.3	2.3	1.4	-12.3	38.2	-13.6	39.0	-14.9	39.8	-16.5	40.8
			-11.7	27.7	2.5	1.2	-13.5	28.5	-14.9	29.2	-16.3	29.9	-18.1	30.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	16.7	6.4	8.1	11.4	16.3	20.8	24.7	26.6	25.8	22.9	17.4	11.4	8.3		
	DBStd	8.35	5.98	5.32	5.20	4.59	3.86	2.88	2.12	2.29	3.06	4.51	4.99	5.50		
	HDD10.0	431	143	91	47	5	0	0	0	0	0	2	41	102		
	HDD18.3	1598	369	288	221	92	21	1	0	0	4	73	213	314		
	CDD10.0	2888	33	38	91	193	334	442	514	491	387	232	84	48		
	CDD18.3	1014	1	2	7	31	98	193	256	233	141	44	6	2		
	CDH23.3	8983	2	14	71	314	916	1786	2442	2027	1042	331	31	7		
	CDH26.7	3566	0	1	8	77	323	769	1109	858	349	69	2	0		
(14) Wind	WSAvg	2.8	2.7	3.0	3.3	3.5	3.0	2.7	2.6	2.3	2.6	2.4	2.5	2.6		
(15) Precipitation	PrecAvg	1412	100	92	105	80	106	135	165	178	170	102	87	92		
	PrecMax	1878	197	256	208	168	209	315	306	370	550	280	216	192		
	PrecMin	906	34	36	37	12	49	48	94	59	22	3	17	21		
	PrecStd	235	41	49	42	37	43	61	57	72	118	72	51	40		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.9	25.8	27.7	31.2	34.2	36.2	37.0	36.2	33.1	31.1	26.3	24.0		
		MCWB	17.7	19.0	17.9	19.9	23.1	24.7	26.2	26.2	24.5	22.9	19.7	18.7		
	2%	DB	21.0	22.4	25.4	28.8	32.2	33.9	34.9	33.9	32.1	28.1	23.9	22.2		
		MCWB	16.6	16.3	17.9	19.4	22.1	24.4	25.9	25.4	24.2	21.6	18.5	17.7		
	5%	DB	18.7	19.9	22.9	27.2	30.2	32.6	33.4	32.6	30.3	27.1	22.2	19.8		
		MCWB	15.2	15.5	16.1	18.5	21.7	24.0	25.4	25.1	23.5	20.9	17.6	17.0		
	10%	DB	16.7	17.7	21.0	25.0	28.0	31.2	32.3	31.3	28.7	25.0	20.1	17.6		
		MCWB	12.8	13.5	15.2	17.5	20.5	23.5	25.0	24.6	22.8	20.1	16.7	14.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.3	19.8	20.4	23.2	24.7	26.8	28.0	27.6	26.0	25.0	21.9	20.6		
		MCDB	21.6	23.9	24.8	26.8	30.1	33.0	34.2	33.0	30.5	28.2	24.4	23.3		
	2%	WB	17.7	18.2	19.0	21.3	23.7	25.8	27.0	26.7	25.2	23.7	20.1	19.0		
		MCDB	19.8	20.9	23.2	25.9	29.2	31.8	32.9	32.2	29.6	26.5	22.6	21.0		
	5%	WB	15.9	16.4	17.7	20.1	22.8	25.0	26.3	25.9	24.5	22.6	18.7	17.4		
		MCDB	18.3	18.9	21.0	24.8	28.2	30.6	32.1	31.0	28.5	25.3	21.1	19.1		
	10%	WB	13.7	14.1	16.4	18.9	21.9	24.3	25.6	25.3	23.7	21.2	17.1	15.4		
		MCDB	16.0	17.5	20.1	23.5	26.8	29.2	30.7	29.8	27.3	24.1	19.5	17.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.6	12.4	13.4	13.5	12.7	11.5	10.8	10.4	10.5	12.1	12.8	11.6		
		MCDBR	13.9	15.0	15.6	15.4	14.2	13.1	12.2	12.0	12.2	12.8	13.8	12.5		
		MCWBR	10.3	10.2	9.0	8.1	6.4	5.4	5.1	5.1	5.5	7.0	9.1	9.2		
	5% WB	MCDBR	12.2	13.6	12.8	12.9	11.8	11.7	11.1	10.7	10.5	9.6	11.2	11.6		
		MCWBR	10.3	10.6	8.7	7.6	5.7	5.4	4.9	4.8	5.3	5.7	8.8	9.7		
(40) Clear-Sky Solar Irradiance	taub		0.318	0.330	0.359	0.397	0.452	0.492	0.525	0.506	0.421	0.375	0.339	0.323		
	taud		2.518	2.479	2.422	2.311	2.201	2.134	2.057	2.105	2.343	2.446	2.502	2.539		
	Ebn at Noon		890	916	913	889	841	803	774	781	838	853	856	862		
	Edh at Noon		82	95	109	128	145	155	167	156	117	98	83	77		
(44) All-Sky Solar Radiation	RadAvg		2.66	3.37	4.45	5.58	6.12	6.37	6.05	5.48	4.57	3.84	2.94	2.35		
	RadStd		0.17	0.33	0.33	0.35	0.43	0.36	0.35	0.32	0.32	0.44	0.22	0.18		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
Regional (44 neighbors)	+0.47	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+159	+115

Nomenclature: See separate page